

## 40.4 Generalized estimating equations

Suppose the question is the marginal one and the cluster structure a nuisance. Then a generalized linear mixed model asks for more than is needed: a distribution for the random effects, a conditional distribution for the responses, and an intractable integral. Everything the question needs is in the *mean*, and [Chapter 38](#) showed that a mean and a variance function suffice to build a usable estimating equation without a likelihood.

Carrying that across to clustered data gives the method of Liang and Zeger (1986), the most widely used in this chapter and the most widely misunderstood. It estimates a marginal coefficient consistently even when the correlation structure it assumes is wrong. It does not estimate a conditional coefficient at all; it produces no likelihood, so most of the model-comparison machinery of the book does not apply; and its robustness to a wrong correlation does not extend to data that go missing for the wrong reasons.

### 40.4.1 The estimating equation

Clusters  $i = 1, \dots, m$  are independent, with response vectors  $\mathbf{Y}_i$  of length  $n_i$  and model matrices  $\mathbf{X}_i$  of size  $n_i \times p$ . The model constrains the marginal mean only:

$$\mathbb{E}(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}), \quad \mu_{ij}(\boldsymbol{\beta}) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta}), \quad (40.4.1)$$

with  $h$  the inverse of a link as in Definition 34.2.2. Nothing is assumed about the joint distribution of  $\mathbf{Y}_i$ , and in particular nothing about its covariance — although a guess at that covariance is used, deliberately.

#### Definition 40.4.1. Generalized estimating equations

Let  $V$  be a variance function and  $\phi$  a dispersion, and write

$$\begin{aligned} \mathbf{A}_i^V(\boldsymbol{\beta}) &= \text{diag}\{V(\mu_{i1}), \dots, V(\mu_{in_i})\}, \\ \mathbf{D}_i(\boldsymbol{\beta}) &= \frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}^\top} = \text{diag}\{h'(\eta_{ij})\} \mathbf{X}_i. \end{aligned}$$

Let  $\mathbf{R}_i(\boldsymbol{\alpha})$  be a **working correlation matrix**: a correlation matrix for the  $n_i$  observations of cluster  $i$ , depending on a finite parameter  $\boldsymbol{\alpha}$ , chosen by the analyst and *not* assumed correct. The **working covariance** is

$$\mathbf{V}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \phi) = \phi (\mathbf{A}_i^V)^{1/2} \mathbf{R}_i(\boldsymbol{\alpha}) (\mathbf{A}_i^V)^{1/2}, \quad (40.4.2)$$

and the **generalized estimating equation** is

$$\mathbf{U}_m(\boldsymbol{\beta}; \boldsymbol{\alpha}, \phi) = \sum_{i=1}^m \mathbf{D}_i(\boldsymbol{\beta})^\top \mathbf{V}_i^{-1} \{\mathbf{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})\} = \mathbf{0}. \quad (40.4.3)$$

In practice  $\boldsymbol{\alpha}$  and  $\phi$  are replaced by moment estimates computed from the Pearson residuals  $r_{ij} = (y_{ij} - \hat{\mu}_{ij}) / \sqrt{V(\hat{\mu}_{ij})}$  at the current  $\hat{\boldsymbol{\beta}}$ , and [Eq. 40.4.3](#) is re-solved, until nothing changes.

Within this chapter  $\mathbf{R}_i$  denotes the working correlation matrix, not the error covariance  $\mathbf{R}$  of

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*Remark* (Three familiar special cases). With  $\mathbf{R}_i = \mathbf{I}$  and one observation per cluster, [Eq. 40.4.3](#) is exactly the likelihood equation [Eq. 34.3.1](#), with  $\mathbf{W}\mathbf{D}^{-1}$  written out. With  $\mathbf{R}_i = \mathbf{I}$  and several observations per cluster it is still that equation, applied as if the observations were independent: the same  $\hat{\boldsymbol{\beta}}$ , different standard errors. And with a general  $\mathbf{R}_i$  but one cluster, it is the quasi-score of [Chapter 38](#) (Theorem 38.2.2) with a full covariance matrix — which for the identity link and  $V \equiv 1$  is the generalized least squares of [Proposition 33.4.1](#).

Solving [Eq. 40.4.3](#) is Fisher scoring. Write  $\mathbf{S}_i = \text{diag}\{h'(\eta_{ij})/\sqrt{V(\mu_{ij})}\}$ ; then  $\mathbf{D}_i^\top \mathbf{V}_i^{-1} = \phi^{-1} \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} (\mathbf{A}_i^V)^{-1/2}$ , so the update is

$$\hat{\boldsymbol{\beta}} \leftarrow \hat{\boldsymbol{\beta}} + \left( \sum_i \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} \mathbf{S}_i \mathbf{X}_i \right)^{-1} \sum_i \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} \mathbf{r}_i, \quad (40.4.4)$$

in which  $\phi$  cancels. It is the iteratively reweighted least squares of [Theorem 34.4.1](#) with  $\mathbf{R}_i^{-1}$  inserted between the weights, and it converges just as quickly.

## 40.4.2 What the estimate does and does not have

### Theorem 40.4.1. Consistency, asymptotic normality and the sandwich

Let the clusters be independent, let  $\boldsymbol{\beta}^0$  be the true value in [Eq. 40.4.1](#), and let  $\boldsymbol{\Sigma}_i^0 = \text{Cov}(\mathbf{Y}_i)$  be arbitrary and unknown. Assume

- (E1)  $n_i \leq n_{\max}$  and  $\|\mathbf{x}_{(ij)}\| \leq C$  for all  $i, j$ ;
- (E2)  $h$  is twice continuously differentiable with  $h'$  bounded,  $V$  is continuous and bounded away from 0 on the relevant range, and  $\boldsymbol{\beta}^0$  is interior to a compact parameter set;
- (E3) the eigenvalues of  $\mathbf{R}_i(\boldsymbol{\alpha})$  are bounded above and away from zero, uniformly in  $i$  and in  $\boldsymbol{\alpha}$  near its limit;
- (E4)  $E\|\mathbf{Y}_i\|^4 \leq K$  for all  $i$ ;
- (E5)  $\hat{\boldsymbol{\alpha}}$  and  $\hat{\phi}$  are  $\sqrt{m}$ -consistent for some limits  $\boldsymbol{\alpha}^*, \phi^*$ , which need not make  $\mathbf{V}_i$  equal to  $\boldsymbol{\Sigma}_i^0$ ;
- (E6) with  $\mathbf{V}_i$  evaluated at  $(\boldsymbol{\beta}^0, \boldsymbol{\alpha}^*, \phi^*)$ ,

$$\mathbf{A}_m = \frac{1}{m} \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{A},$$

$$\mathbf{B}_m = \frac{1}{m} \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \boldsymbol{\Sigma}_i^0 \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{B},$$

both positive definite.

Then:

- (a) **Unbiasedness.** For every fixed  $\boldsymbol{\alpha}$  and  $\phi$ ,  $E\{\mathbf{U}_m(\boldsymbol{\beta}^0; \boldsymbol{\alpha}, \phi)\} = \mathbf{0}$ . The estimating function is unbiased whatever the working correlation is.
- (b) **Consistency and asymptotic normality.** With probability tending to one, [Eq. 40.4.3](#) has a root  $\hat{\boldsymbol{\beta}}_m$  with  $\hat{\boldsymbol{\beta}}_m \rightarrow \boldsymbol{\beta}^0$  in probability, and

$$\sqrt{m}(\hat{\boldsymbol{\beta}}_m - \boldsymbol{\beta}^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}). \quad (40.4.5)$$

(c) **The sandwich estimator.** Let  $\mathbf{r}_i = \mathbf{Y}_i - \boldsymbol{\mu}_i(\hat{\boldsymbol{\beta}}_m)$  and

$$\begin{aligned} \hat{\mathbf{V}}_R &= \hat{\mathbf{A}}_m^{-1} \left( \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i \mathbf{r}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right) \hat{\mathbf{A}}_m^{-1}, \\ \hat{\mathbf{A}}_m &= \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i, \end{aligned} \quad (40.4.6)$$

with hats denoting evaluation at  $\hat{\boldsymbol{\beta}}_m, \hat{\boldsymbol{\alpha}}, \hat{\phi}$ ;  $\hat{\mathbf{A}}_m$  is  $m$  times the sample version of the  $\mathbf{A}_m$  of (E6). Then  $m\hat{\mathbf{V}}_R \rightarrow \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  in probability, and the Wald statistic of a hypothesis  $\mathbf{A}^\top \boldsymbol{\beta}$  of rank  $q$ , formed with  $\hat{\mathbf{V}}_R$ , tends to  $\chi^2(q)$ .

(d) **When the model-based covariance is valid.** If the working covariance is correct,  $\mathbf{V}_i = \boldsymbol{\Sigma}_i^0$ , then  $\mathbf{B} = \mathbf{A}$  and Eq. 40.4.5 reduces to  $\mathbf{A}^{-1}$ , which the **model-based** estimator  $\hat{\mathbf{V}}_M = (\sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i)^{-1}$  estimates consistently. Otherwise  $\hat{\mathbf{V}}_M$  is inconsistent, in either direction.

*Proof.*

- (a) At  $\boldsymbol{\beta} = \boldsymbol{\beta}^0$  the matrices  $\mathbf{D}_i$  and  $\mathbf{V}_i$  are deterministic — they depend on the data only through  $\boldsymbol{\beta}$ , which is held fixed — so  $\mathbf{E} \mathbf{U}_m = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{E}\{\mathbf{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}^0)\} = \mathbf{0}$  by Eq. 40.4.1. Nothing about  $\mathbf{R}_i$  enters, and that is the whole source of the method's robustness.
- (b) and (c). Expand  $\mathbf{U}_m(\boldsymbol{\beta}; \boldsymbol{\alpha})$  about  $\boldsymbol{\beta}^0$  at  $\boldsymbol{\alpha} = \hat{\boldsymbol{\alpha}}$ . Since  $\mathbf{U}_m$  is vector-valued, no single intermediate point serves, so we use the integral form of Taylor's theorem:

$$\begin{aligned} \mathbf{0} &= \mathbf{U}_m(\hat{\boldsymbol{\beta}}_m; \hat{\boldsymbol{\alpha}}) = \mathbf{U}_m(\boldsymbol{\beta}^0; \hat{\boldsymbol{\alpha}}) + \bar{\mathbf{J}}_m (\hat{\boldsymbol{\beta}}_m - \boldsymbol{\beta}^0), \\ \bar{\mathbf{J}}_m &= \int_0^1 \frac{\partial \mathbf{U}_m}{\partial \boldsymbol{\beta}^\top} (\boldsymbol{\beta}^0 + t(\hat{\boldsymbol{\beta}}_m - \boldsymbol{\beta}^0)) dt. \end{aligned}$$

Three facts give the result.

*The derivative.*  $\partial \mathbf{U}_m / \partial \boldsymbol{\beta}^\top = -\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i + \mathbf{T}_m$ , where  $\mathbf{T}_m$  collects the terms in which the derivative falls on  $\mathbf{D}_i^\top \mathbf{V}_i^{-1}$  and therefore still multiplies  $\mathbf{Y}_i - \boldsymbol{\mu}_i$ . Each summand of  $\mathbf{T}_m$  has mean zero at  $\boldsymbol{\beta}^0$  and bounded variance by (E1), (E2) and (E4), so  $m^{-1}\mathbf{T}_m \rightarrow \mathbf{0}$  in probability by the weak law for independent summands; and  $m^{-1}\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{A}$  by (E6). Hence  $m^{-1}\partial \mathbf{U}_m / \partial \boldsymbol{\beta}^\top \rightarrow -\mathbf{A}$ , uniformly for  $\boldsymbol{\beta}$  in a neighbourhood of  $\boldsymbol{\beta}^0$  under (E2); the convergence being uniform over that neighbourhood,  $m^{-1}\bar{\mathbf{J}}_m \rightarrow -\mathbf{A}$  as well, once  $\hat{\boldsymbol{\beta}}_m$  is in it.

*The score.*  $m^{-1/2}\mathbf{U}_m(\boldsymbol{\beta}^0; \boldsymbol{\alpha}^*)$  is  $m^{-1/2}$  times a sum of  $m$  independent mean-zero vectors  $\mathbf{s}_i = \mathbf{D}_i^\top \mathbf{V}_i^{-1}(\mathbf{Y}_i - \boldsymbol{\mu}_i)$  with  $\sum_i \text{Cov}(\mathbf{s}_i) = m\mathbf{B}_m$ . Conditions (E1)–(E4) bound  $\mathbf{E}\|\mathbf{s}_i\|^4$  uniformly, which gives the Lyapunov condition, so the Lindeberg–Feller theorem applies and  $m^{-1/2}\mathbf{U}_m \rightarrow N_p(\mathbf{0}, \mathbf{B})$ .

*Estimating  $\boldsymbol{\alpha}$  does not matter.* Differentiating Eq. 40.4.3 with respect to  $\boldsymbol{\alpha}$  leaves the factor  $\mathbf{Y}_i - \boldsymbol{\mu}_i$  untouched, so  $\partial \mathbf{U}_m / \partial \boldsymbol{\alpha}^\top$  at  $\boldsymbol{\beta}^0$  is again a sum of mean-zero terms and  $m^{-1}\partial \mathbf{U}_m / \partial \boldsymbol{\alpha}^\top \rightarrow \mathbf{0}$ . Therefore  $m^{-1/2}\mathbf{U}_m(\boldsymbol{\beta}^0; \hat{\boldsymbol{\alpha}}) = m^{-1/2}\mathbf{U}_m(\boldsymbol{\beta}^0; \boldsymbol{\alpha}^*) + \{m^{-1}\partial \mathbf{U}_m / \partial \boldsymbol{\alpha}^\top\} \sqrt{m}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = m^{-1/2}\mathbf{U}_m(\boldsymbol{\beta}^0; \boldsymbol{\alpha}^*) + o_p(1)$  by (E5).

Combining,  $\sqrt{m}(\hat{\boldsymbol{\beta}}_m - \boldsymbol{\beta}^0) = \mathbf{A}^{-1}m^{-1/2}\mathbf{U}_m(\boldsymbol{\beta}^0) + o_p(1)$ , which is Eq. 40.4.5. For (c),  $\mathbf{r}_i \mathbf{r}_i^\top$  has mean  $\boldsymbol{\Sigma}_i^0 + O(m^{-1})$  at  $\hat{\boldsymbol{\beta}}_m$ , and the argument of Theorem 21.4.2 with the clus-

ter in place of the observation — the version already used for Proposition 33.4.1(b) — gives  $m^{-1} \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i \mathbf{r}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \rightarrow \mathbf{B}$ . The Wald limit follows from Eq. 40.4.5 and the continuous mapping theorem.

(d) If  $\mathbf{V}_i = \Sigma_i^0$  then  $\mathbf{D}_i^\top \mathbf{V}_i^{-1} \Sigma_i^0 \mathbf{V}_i^{-1} \mathbf{D}_i = \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$  for every  $i$ , so  $\mathbf{B}_m = \mathbf{A}_m$ . ■

*Remark* (What is proved and what is assumed). The existence of a consistent root and the uniform control of the derivative are the two places where the argument leans on standard machinery rather than its own steam; both go exactly as for Theorem 34.5.1, using the concavity-free argument for estimating equations in Liang and Zeger (1986). Condition (E5) is a real assumption: moment estimators of  $\boldsymbol{\alpha}$  from Pearson residuals are  $\sqrt{m}$ -consistent under (E1)–(E4), but an unstructured  $\mathbf{R}$  estimated from few clusters is not. Note the notation:  $\mathbf{U}_m$  is the estimating *function*, as in Chapter 38, while a hatted  $\mathbf{V}$  with a letter subscript is a *covariance estimator* for  $\hat{\boldsymbol{\beta}}$ , as in Chapter 21;  $\mathbf{V}_i$ , with a cluster subscript, remains the working covariance of cluster  $i$ .

*Warning* (The mean model must hold for the whole cluster). Eq. 40.4.1 constrains  $E(\mathbf{Y}_i)$ , the mean of the whole cluster vector, and part (a) uses it in that form; what one usually has in mind is the weaker  $E(Y_{ij} | \mathbf{x}_{(ij)}) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta})$ . A *non-diagonal*  $\mathbf{R}_i$  mixes observations across occasions, so unbiasedness then needs the **full-covariate conditional mean** condition  $E(Y_{ij} | \mathbf{X}_i) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta})$ : the mean at occasion  $j$  must not depend on the covariates at other occasions. With time-varying covariates and feedback — this year’s investment changing next year’s market value, a dose adjusted in response to last month’s reading — that fails even though the occasion-by-occasion mean model is correct, and a non-diagonal working correlation is then *inconsistent* while independence GEE is not (Pepe and Anderson 1994; Diggle, Heagerty, Liang and Zeger 2002, chapter 12).

*Idea*. The asymptotics are in  $m$ , the number of clusters; the total sample size is irrelevant. A trial with ten thousand patients in six hospitals has six independent pieces of information about anything that varies between hospitals, and the sandwich cannot manufacture more — the lesson of Section 33.2.

### 40.4.3 Travel-mode choice

#### Example. Four options per traveller

Each of 210 travellers between Australian cities chose one of four modes — air, train, bus, car — and the data record, for every traveller and every mode, the terminal waiting time and a generalized cost. Take  $Y_{ij} = 1$  if traveller  $i$  chose mode  $j$ , fit a logistic model with mode indicators, waiting time and cost, and treat each traveller as a cluster of  $n_i = 4$ . The sample deliberately over-represents the rarer modes, so by Theorem 35.5.1 the mode indicators and the intercept are shifted by the sampling design and are not population quantities; the within-traveller coefficients are unaffected, and standard errors are the point here in any

case.

The four records of a traveller are anything but independent: exactly one is a 1. An independence working correlation gives the same coefficients as ordinary logistic regression, so the question is only what the standard errors should be. The ratio of the sandwich standard error to the model-based one ranges from 0.694 to 1.558 across the six coefficients: for terminal waiting time, whose coefficient is  $-1.0693$  per ten minutes, the sandwich standard error is 1.558 times larger (0.1698 against 0.1090); for generalized cost, whose coefficient is  $-0.0098$  per ten dollars, it is 0.694 times as large (0.0138 against 0.0199).

The sandwich is not a device for making standard errors bigger. Negative within-cluster correlation makes *within-cluster contrasts* better determined than independence would suggest, and cost is such a contrast; waiting time is confounded with the mode indicators, which the traveller's single choice constrains, and is therefore worse determined. Two further observations. The Pearson dispersion estimate is 5.155, which looks like extreme overdispersion and means nothing: Bernoulli responses carry no information about a dispersion parameter (Proposition 40.2.1(b) with  $N_i = 1$ ). And the exchangeable working correlation estimate is  $-0.3015$ , close to the smallest value  $-1/(n_i - 1) = -1/3$  an exchangeable matrix of size four can have: there the smallest eigenvalue of  $\mathbf{R}$  is  $1 + 3\hat{\alpha} = 0.0956$ . An almost singular structure is a bad working covariance, and independence is the sensible choice.

```
import numpy as np

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

def working_correlation(kind, r, p, scale_fixed=False):
    """Moment estimate of the working correlation from the Pearson residuals r (m by n)."""
    m, n = r.shape
    phi = 1.0 if scale_fixed else np.sum(r**2) / (m * n - p)
    if kind == "independence":
        return np.eye(n), 0.0, phi
    if kind == "exchangeable":
        off = (np.sum(r.sum(1)**2) - np.sum(r**2)) / 2
        a = off / (phi * (m * n * (n - 1) / 2 - p))
        return np.eye(n) + a * (1 - np.eye(n)), a, phi
    if kind == "ar1":
        a = np.sum(r[:, :-1] * r[:, 1:]) / (phi * (m * (n - 1) - p))
        lag = np.abs(np.arange(n)[:, None] - np.arange(n))
        return a**lag, a, phi
    if kind == "unstructured":
        R = (r.T @ r) / (phi * m)
        R = R / np.sqrt(np.outer(np.diag(R), np.diag(R)))
        return R, np.nan, phi
    raise ValueError(kind)

def gee(y, X, kind="independence", family="binomial", scale_fixed=False, maxit=200):
    """Solve  $\sum_i D_i' V_i^{-1} (y_i - \mu_i) = 0$  for balanced clusters y (m, n), X (m, n, p).

    With  $S_i = \text{diag}(\mu_{ij}' / \sqrt{V(\mu_{ij})})$  and Pearson residuals  $r_i$ , the Fisher-scoring step is  $\beta \leftarrow (\sum_i X_i' S_i R^{-1} S_i X_i)^{-1} \sum_i X_i' S_i R^{-1} r_i$ ."""
    m, n, p = X.shape
    beta, R, alpha, phi = np.zeros(p), np.eye(n), 0.0, 1.0
```

```

for _ in range(maxit):
    eta = X @ beta
    if family == "binomial":
        mu = expit(eta)
        dmu, V = mu * (1 - mu), mu * (1 - mu)
    else:
        # log link,  $V(\mu) = \mu^2 (1/\gamma)$ 
        mu = np.exp(eta)
        dmu, V = mu, mu**2
    r = (y - mu) / np.sqrt(V)
    R, alpha, phi = working_correlation(kind, r, p, scale_fixed)
    Rinv = np.linalg.inv(R)
    S = dmu / np.sqrt(V)
    SX = S[..., None] * X
    # (m, n, p)
    A = np.einsum("mjp,jk,mkq->pq", SX, Rinv, SX)
    g = np.einsum("mjp,jk,mk->p", SX, Rinv, r)
    step = np.linalg.solve(A, g)
    beta = beta + step
    if np.max(np.abs(step)) < 1e-11:
        break
u = np.einsum("mjp,jk,mk->mp", SX, Rinv, r)
# cluster contributions
Ainv = np.linalg.inv(A)
robust = Ainv @ (u.T @ u) @ Ainv
# the sandwich covariance
naive = phi * Ainv
# the model-based covariance
return dict(beta=beta, naive=naive, robust=robust, alpha=alpha, phi=phi, R=R,
            A=A, Ainv=Ainv, S=S, r=r, Rinv=Rinv, X=X)

```

[Run this code online](#)

```

import pandas as pd
import statsmodels.api as sm

d = sm.datasets.modechoice.load_pandas().data.sort_values(["individual", "mode"])
m, n = d["individual"].nunique(), 4
# 210 travellers, 4 modes each
y = d["choice"].to_numpy().reshape(m, n)
D = pd.get_dummies(d["mode"].astype(int), drop_first=True).to_numpy(dtype=float)
cols = [np.ones(m * n)] + [D[:, j] for j in range(3)] + [d["ttme"] / 10, d["gc"] / 10]
X = np.stack([np.asarray(c, dtype=float).reshape(m, n) for c in cols], axis=2)

# a Bernoulli response carries no information about a dispersion parameter, so phi = 1
fit = gee(y, X, "independence", scale_fixed=True)
se_naive, se_robust = np.sqrt(np.diag(fit["naive"])), np.sqrt(np.diag(fit["robust"]))
names = ["intercept", "train", "bus", "car", "ttme/10", "gc/10"]
for k, name in enumerate(names):
    print(f"{name:10s} {fit['beta'][k]:8.4f} naive {se_naive[k]:.4f} robust {se_robust[k]:.4f}")
print("exchangeable alpha:", round(gee(y, X, 'exchangeable', scale_fixed=True)['alpha'], 4))

```

[Run this code online](#)

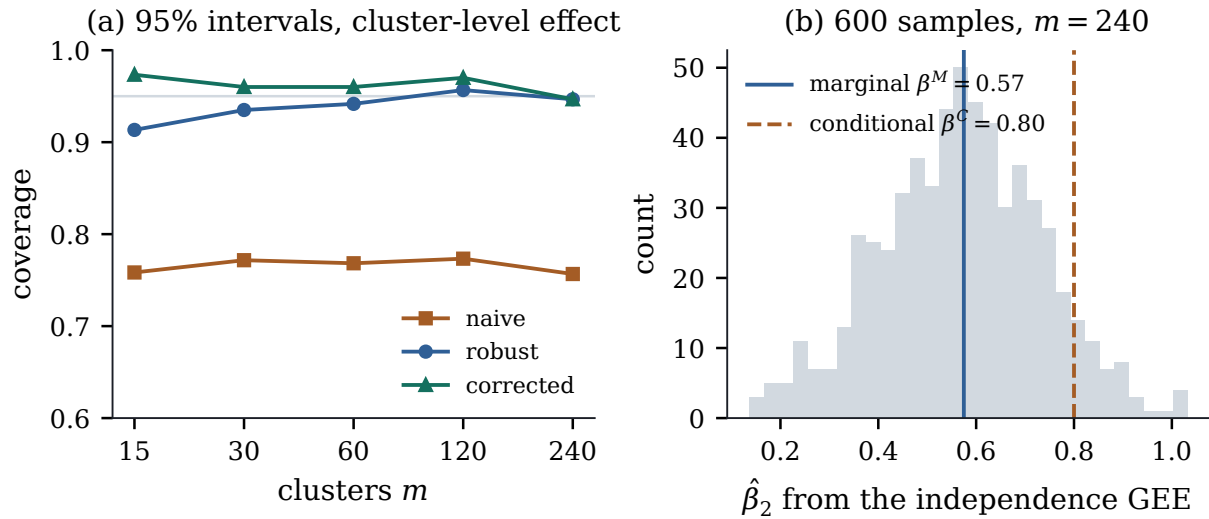


Figure 40.4.1: GEE with a wrong working correlation. (a) Coverage of nominal 95% intervals for the marginal coefficient of a cluster-level covariate, against the number of clusters, for three covariance estimators. (b) The sampling distribution of  $\hat{\beta}_2$  at  $m = 240$ , against the marginal and conditional values.

#### 40.4.4 Small samples, and a simulation

The sandwich Eq. 40.4.6 replaces  $\Sigma_i^0$  by  $\mathbf{r}_i \mathbf{r}_i^\top$ , and the residuals are too small because  $\hat{\beta}_m$  was fitted to them; the bias is of order  $p/m$  and downward, exactly as for Theorem 21.4.2 and Exercise (C2). Two standard repairs, both used in [Example \(Coverage when the working correlation is wrong\)](#):

- **Bias-corrected residuals.** Mancl and DeRouen (2001) replace  $\mathbf{r}_i$  by  $(\mathbf{I} - \mathbf{H}_i)^{-1} \mathbf{r}_i$ , with  $\mathbf{H}_i = \mathbf{D}_i (\sum_k \mathbf{D}_k^\top \mathbf{V}_k^{-1} \mathbf{D}_k)^{-1} \mathbf{D}_i^\top \mathbf{V}_i^{-1}$  cluster  $i$ 's leverage block — the clustered analogue of the HC3 correction of Definition 21.4.1. Kauermann and Carroll (2001) use  $(\mathbf{I} - \mathbf{H}_i)^{-1/2}$ , which corrects less.
- **A  $t$  reference.** Replace the normal quantile by  $t_{m-p}$ . With  $m = 15$  and  $p = 3$  this widens a 95% interval by  $t_{12,0.975}/z_{0.975} = 2.179/1.960$ , about eleven per cent.

##### Example. Coverage when the working correlation is wrong

Generate clusters of  $n_i = 8$  binary responses from a random-intercept logistic model with  $\tau = 1.5$ , one covariate varying within the cluster and one fixed for the whole cluster (a treatment assigned to clusters), and fit the marginal model by GEE with an *independence* working correlation — wrong by construction. The target is the marginal cluster-level coefficient, which Eq. 40.1.3 puts at 0.5748 when the conditional coefficient is 0.8. Over 600 replications at  $m = 240$  the mean estimate is 0.5681: consistent for the marginal coefficient, and not for the conditional one.

The three intervals behave quite differently. The model-based interval covers 0.757 of the time and does not improve with  $m$ : it is wrong, not merely imprecise. The sandwich covers 0.913 at  $m = 15$  and 0.947 at  $m = 240$ , and the bias-corrected sandwich with a  $t$  reference covers 0.973 at  $m = 15$  — slightly conservative, which at that sample size is the right side to err on.

## 40.4.5 What GEE does not give

*No likelihood.* Eq. 40.4.3 is not the derivative of anything in general, so there is no deviance, no likelihood ratio test, and no AIC or BIC. Nested mean models are compared by Wald tests built on Eq. 40.4.6 or by score-type tests built on Eq. 40.4.3 under the null; selection uses QIC, the quasi-likelihood analogue of AIC (Section 40.5). Reporting a “deviance” from GEE output is a category error.

*No random effects and no distribution.* There is nothing to predict, so if the object of interest is a cluster — which hospital is doing badly, which firm is unusually profitable — GEE has no answer and Proposition 40.2.3 has no analogue; and because only the mean is modelled, the fit cannot be simulated from, cannot give a prediction interval, and cannot be checked by an envelope or posterior predictive method.

*Few clusters.* Everything above is asymptotic in  $m$ . Below about forty clusters the sandwich needs correcting, and below about fifteen no correction reliably saves it; a likelihood-based method with a covariance model from Definition 33.4.1 is then the better bet.

## 40.4.6 Missing data

The robustness of Eq. 40.4.3 has a sharp limit. If some responses are not observed, write  $R_{ij} = 1$  when  $Y_{ij}$  is recorded, so that the estimating function is really  $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{R}_i (\mathbf{Y}_i - \boldsymbol{\mu}_i)$  with  $\mathbf{R}_i = \text{diag}(R_{ij})$ . Part (a) of Theorem 40.4.1 needs  $E\{\mathbf{R}_i (\mathbf{Y}_i - \boldsymbol{\mu}_i)\} = \mathbf{0}$ , a genuine restriction on the missingness.

If the data are **missing completely at random**,  $\mathbf{R}_i$  is independent of  $\mathbf{Y}_i$ , the expectation factorizes, and everything above holds unchanged; the same is true if missingness depends only on covariates that are in the model and fully observed, because the mean model is then still correct conditionally on being observed. But if missingness depends on *previously observed responses* — a patient who was doing badly last visit is more likely to drop out — then  $\mathbf{R}_i$  and  $\mathbf{Y}_i - \boldsymbol{\mu}_i$  are dependent, the estimating function is biased and GEE is inconsistent. This is the **missing at random** case of Definition 41.1.1, and it is exactly the case in which a likelihood-based analysis remains valid.

That asymmetry is the strongest practical argument for the mixed-model route in longitudinal studies with dropout, and it is not widely enough known. The repair is to weight each cluster’s contribution by the inverse of its estimated probability of being observed — the weighted GEE of Robins, Rotnitzky and Zhao (1995), which trades the assumption about the correlation for one about the missingness model. Chapter 41 develops the mechanisms, the ignorability theorem and the alternatives.

## 40.4.7 Exercises

### 40.4.7.1 A. Check your understanding

#### Exercise. A1

Show from Eq. 40.4.3 that with  $\mathbf{R}_i = \mathbf{I}$  the estimate  $\hat{\boldsymbol{\beta}}$  does not depend on  $\phi$  or on the cluster structure at all, and equals the ordinary generalized linear model estimate. Which quantities *do* depend on the clustering?

**Exercise. A2**

A trial randomizes clinics to a new protocol and measures a binary outcome on patients within clinics. The investigators want to report “the effect of the protocol on a patient’s odds of recovery”. Which method should they use, and what would a GEE analysis report instead?

**40.4.7.2 B. Practice****Exercise. B1**

Take  $m$  clusters of size 2, the identity link,  $V \equiv 1$ ,  $\phi = 1$ , and an exchangeable working correlation with parameter  $\alpha$ . Show that Eq. 40.4.3 reduces to weighted least squares with weight matrix  $\mathbf{R}^{-1}$ ,  $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{J}$ , in each cluster, and that when  $\mathbf{x}_{(i1)} = \mathbf{x}_{(i2)}$  for every  $i$  the estimate does not depend on  $\alpha$  at all. Interpret.

*Solution.* The estimating equation is  $\sum_i \mathbf{X}_i^\top \mathbf{R}^{-1}(\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) = \mathbf{0}$ , which is the normal equation of generalized least squares with covariance  $\mathbf{R}$ . If the two rows of  $\mathbf{X}_i$  are equal,  $\mathbf{X}_i = \mathbf{1}\mathbf{x}_{(i)}^\top$  and  $\mathbf{X}_i^\top \mathbf{R}^{-1} \mathbf{X}_i = (\mathbf{1}^\top \mathbf{R}^{-1} \mathbf{1})\mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$ ,  $\mathbf{X}_i^\top \mathbf{R}^{-1} \mathbf{y}_i = \mathbf{x}_{(i)} \mathbf{1}^\top \mathbf{R}^{-1} \mathbf{y}_i$ . For an exchangeable  $\mathbf{R}$ ,  $\mathbf{R}^{-1} \mathbf{1} = \{1/(1 + \alpha)\}\mathbf{1}$ , so both sides carry the same scalar factor  $1/(1 + \alpha)$  and it cancels. A cluster-constant covariate sees only the cluster mean, and reweighting a mean by an exchangeable covariance does nothing.

**Exercise. B2**

Let  $\mathbf{G}_i$  be any  $n_i \times p$  matrix depending on  $\boldsymbol{\beta}$  but not on  $\mathbf{Y}_i$ . Show that  $\sum_i \mathbf{G}_i^\top (\mathbf{Y}_i - \boldsymbol{\mu}_i)$  is an unbiased estimating function and that Eq. 40.4.3 is the member with  $\mathbf{G}_i = \mathbf{V}_i^{-1} \mathbf{D}_i$ . Which member is optimal, in the sense of Proposition 38.2.3, and why is it not available?

*Solution.* Unbiasedness is immediate from  $E(\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ . The Godambe-optimal choice within this linear class is  $\mathbf{G}_i = (\boldsymbol{\Sigma}_i^0)^{-1} \mathbf{D}_i$ , the *true* covariance in place of the working one; Proposition 38.2.3 is the same statement with  $n_i = 1$ . It is unavailable because  $\boldsymbol{\Sigma}_i^0$  is unknown, which is precisely what makes a working correlation necessary.

**40.4.7.3 C. Going deeper****Exercise. C1**

Show that if  $\mathbf{V}_i = \boldsymbol{\Sigma}_i^0$  then  $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \mathbf{A}^{-1}$  is the smallest asymptotic covariance attainable in the class of Exercise (B2), in the nonnegative-definite ordering. (Hint: for any  $\mathbf{G}_i$ , compare  $(\sum \mathbf{G}_i^\top \mathbf{D}_i)^{-1} (\sum \mathbf{G}_i^\top \boldsymbol{\Sigma}_i^0 \mathbf{G}_i) (\sum \mathbf{D}_i^\top \mathbf{G}_i)^{-1}$  with  $\mathbf{A}^{-1}$  using a Cauchy–Schwarz argument in the inner product  $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_i \mathbf{a}_i^\top \boldsymbol{\Sigma}_i^0 \mathbf{b}_i$ .)